

Habitable-zone Earths at the detection frontier: Measured completeness and false-alarm rate of a transit pipeline for the PLATO M-dwarf sample

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Abstract

The PLATO mission will observe its first long-pointing field, LOPS2 (Long-duration Observation Phase, South field), for at least two years, bringing the habitable zones (HZ) of its M-dwarf sample within reach of a transit search. The completeness and false-alarm rate of a pipeline validated on TESS M dwarfs are measured on simulated PLATO photometry and converted into a candidate-level forecast for the LOPS2 M dwarfs. Simulated PLATO light curves (717 d baseline; three stellar cells of the faint half of the sample) were processed end to end in two arms, without and with a pre-launch model of the instrument systematics: 930 in the first (465 realisations, 525 injected planets, 465 noise-only twins) and 790 in the second. Injected planets measure completeness against the expected signal-to-noise ratio (S/N), separately for the long-period regime of the HZ; noise-only twins measure the false-alarm rate against the floor on the red-noise-aware statistic S/N_{pink} . Completeness reaches 50% at $S/N = 7.7 \pm 0.15$ over all periods and 8.6 ± 0.3 in the long-period regime. The false-alarm rate falls from $79 \pm 2\%$ to 0.6% as the S/N_{pink} floor rises from 6 to 8, and $6 \pm 1\%$ at the adopted floor 7.5 (completeness $54 \pm 2\%$). In the field, the 50 known transiting objects of interest (19 confirmed planets, 31 candidates) would all be recovered; a transiting HZ planet of $1.0(1.5)R_{\oplus}$ would be recovered as a candidate with $\sim 40\%$ (65%) probability, giving about 4 HZ Earths and 7 super-Earths over 16 929 field M dwarfs. The systematics arm shifts the 50% point by $+0.54^{+0.27}_{-0.22}$ in S/N, a penalty localised on the transition, leaving the floor robust and negligible on the long-period injections that feed the HZ. Habitable-zone Earths around the PLATO M dwarfs sit at the measured detection frontier, where completeness and reliability, both measurable before launch, decide the yield.

Keywords. planets and satellites: detection; planets and satellites: terrestrial planets; techniques: photometric; stars: low-mass

1 Introduction

The PLANetary Transits and Oscillations of stars mission (PLATO; Rauer et al. 2025) will observe its first long-pointing field, LOPS2 (Long-duration Observation Phase, South field; Nascimbeni et al. 2025), continuously for at least two years. Its cool-dwarf sample (P4; Prisinzano et al. 2026) extends the transit search to M dwarfs, for which the habitable zone (HZ) corresponds to orbital periods of 19–112 d for M4 to M2 hosts; the 717 d baseline therefore accumulates 6–37 HZ transits per target.

Yield estimates for PLATO exist along three lines: injection and recovery of Earth-like planets around Sun-like stars (Heller et al. 2022), mission-wide statistical yields for the FGK samples (Matuszewski et al. 2023), and demographic survey simulations that reach the M dwarfs (Schlecker et al. 2024). The present work complements these forecasts with the two quantities that injection and recovery through a search-and-screening pipeline can provide: the fraction of injected planets the pipeline returns as candidates from realistic light curves, and the rate at which the same pipeline returns a candidate from light curves that contain no planet. This paper reports both for the cool-dwarf sample; the quoted yields are candidate-level, before the astrophysical false-positive vetting that requires

imaging or spectroscopy.

The pipeline was developed and validated on M dwarfs observed by the Transiting Exoplanet Survey Satellite (TESS; Tschudi 2026a). Applied uniformly to the 461 M-dwarf hosts of TESS objects of interest (TOIs) listed by the Exoplanet Follow-up Observing Program (ExoFOP), it re-detected 165 of the 193 confirmed planets in its period range (85.5%) and misclassified none of them as a false positive (Tschudi 2026b).

2 Simulated campaign and pipeline

Light curves were generated with the PLATO Solar-like Light-curve Simulator (PSLS; Samadi et al. 2019) at the 25 s cadence of the P4 data products (Rauer et al. 2025) and binned to 300 s for the search. The binning divides the search cost by an order of magnitude (2×10^5 instead of 2.5×10^6 points per 717 d light curve) at a transit S/N cost below 1%, the bins remaining much shorter than any P4 transit duration. The native 25 s cadence remains available for the per-candidate vetting.

The simulated signal contains photon and detector noise, the nominal gap structure, and a rotational modulation from three spots of finite lifetime; the second arm adds the residual systematics described below. Granulation and the

Table 1. Campaign cells, shared by both arms: content and purpose.

Cell	Host	Noise ^(a)	Content (injections + twins)	Measures
M4 bright	M4, $V = 14.5$, 12 cam	1308	145 single planets + 145	completeness transition at the low-noise anchor; HZ band 19–52 d
M2	M2, $V = 15.0$, 12 cam	2076	145 single planets + 145	earlier-type host: HZ band 43–112 d, the long-period extreme (6–17 events)
M4 faint	M4, $V = 16.0$, 6 cam	6200	145 single planets + 145	faint six-camera limit; anchors the noise model at the hard end
L 98-59 analogue	M4, $V = 14.5$, 12 cam	1308	10×3 planets + 10	iterative multiplanet search; outer pair at a 2.02:1 ratio (harmonic confusion)
TOI-700 analogue	M2, $V = 15.0$, 12 cam	2076	10×4 planets + 10	long-period multiplicity up to the HZ planet of the real system
Two-planet	M4, $V = 14.5$, 12 cam	1308	10×2 planets + 10	near 3:2 pair; mutual masking of close signals

Notes. ^(a) ppm per 300 s bin. Single-planet injection periods span 0.5–100 d. The multiplanet periods are 2.25/3.69/7.45 d (L 98-59 analogue), 10.0/16.1/27.8/37.4 d (TOI-700 analogue), and 2.44/3.79 d (two-planet).

stochastic activity term were disabled, being calibrated on solar-like and evolved stars, and solar-like oscillations were disabled because the modes of main-sequence M dwarfs lie far above the 1667 μHz Nyquist frequency of the 300 s bins. Flares were not simulated, the most consequential omission for an M-dwarf sample (Sect. 5).

Three stellar cells anchor the campaign: an M4 dwarf at $V = 14.5$ observed with 12 cameras (1308 ppm per 300 s bin), an M2 at $V = 15.0$ with 12 cameras (2076 ppm), and an M4 at $V = 16.0$ with 6 cameras (6200 ppm). These cells span the faint half of the sample. The median magnitude of the field M dwarfs is $V = 15.5$; at these magnitudes the expected S/N of HZ planets of $1\text{--}1.5 R_{\oplus}$ falls on the completeness transition, where recovery drops from likely to unlikely, while brighter stars sit on the saturated plateau. The injections therefore concentrate on the transition, as in the TESS survey at the stellar noise frontier (Tschudi 2026a).

Throughout, and for the field stars of Sect. 4, the expected S/N is $\delta_{\text{win}} \sqrt{n_{\text{in}}} / \sigma$, with σ the white noise per 300 s point of the cell, $n_{\text{in}} = (717 \text{ d}/P)(T_{14}/300 \text{ s})$ the number of in-transit bins, and δ_{win} the limb-darkened window depth, which exceeds the geometric $(R_p/R_{\star})^2$ by a measured factor 1.20; using the geometric depth would displace the axis by 20%. The 465 planet-hosting realisations carry 525 injected planets with expected S/N concentrated on that transition (3.5–13); three multiplanet configurations complete the campaign. Each realisation has a noise-only twin, the same star and noise class without a planet; the 465 twins carry the entire false-alarm measurement. Table 1 summarises the cells, their content, and the quantity each one measures; the systematics arm reuses the same cells with fewer trials.

The campaign comprises two arms, matched in cell design, S/N sampling, and chain (435 and 371 single-planet trials, respectively). The arm reported here (arm A) contains no residual instrument systematics and measures the intrinsic performance of the pipeline against stellar and photon noise; the systematics arm (arm B) injects the pre-launch model of the post-correction residuals (Sect. 5) and measures their cost. The two arms share every input except

the systematics, so their difference is the instrumental cost, and the pair reads the pipeline’s intrinsic performance and the instrumental penalty separately.

The pipeline is that of the TESS survey (Tschudi 2026a,b). The chain is: Gaussian-process detrending; an iterative transit-least-squares (TLS) search (Hippke & Heller 2019), whose internal acceptance requires a signal detection efficiency $\text{SDE} \geq 7$; harmonic-alias resolution; and an event-time-coherence test that rejects signals whose per-transit timing is incoherent (window-function aliases of stellar variability). Three settings differ from the published TESS configuration.

First, the search noise is estimated by TLS, the PSLs uncertainties being photon-only. Second, a veto rejects candidates within 0.5% of the quarterly rotation comb (90.004 d and low harmonics), the PLATO analogue of the 13.7 d entry used for TESS: the rotations imprint structures repeating at the quarter period, an instrumental clock rather than a transit. It shapes the false-alarm rate below and costs three injections, none recovered, one at $\text{S/N} = 11.8$ and $P = 29.90$ d that the chain would otherwise have detected; the comb crosses the HZ band of the M2 cell, leaving three narrow blind periods. Third, a power gate on the event-time test lets it abstain rather than reject when the median per-event depth significance falls below 0.85σ , a threshold inside an interval left empty by the calibration set (noise twins reach 0.39, false rejections 0.61, confirmed planets begin at 1.13, and the certified window-function false positive at 1.18 stays rejected). The gate changes 29 of 435 verdicts, all at $P \leq 10.6$ d and none in the HZ band, where recovery is 37 of 75 with and without it, so the HZ forecast is independent of it. Every scored candidate carries an S/N computed from the white and red noise contributions at the transit timescale, noted S/N_{pink} after its pink-/red-noise formulation (Pont et al. 2006); the operating points reported below apply a floor on this statistic.

3 Completeness and false-alarm rate

Figure 1 presents the completeness measurement. Over all periods the recovery fraction reaches 50% at an expected

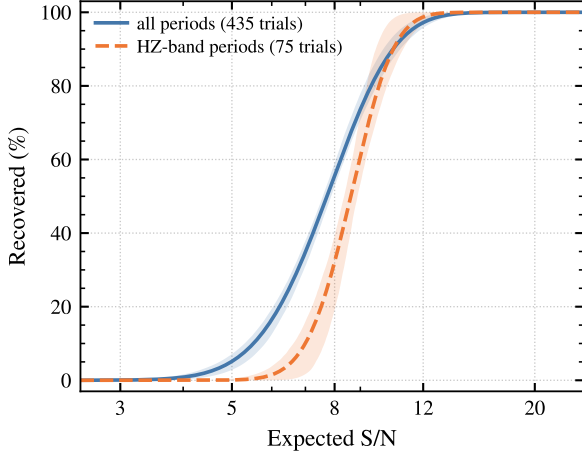


Fig. 1. End-to-end completeness versus expected S/N (no-systematics arm). Blue: all periods (435 planets; the fits use the single-planet injections only, the 90 planets of the multiplanet configurations being fixed-period architecture tests, see Table 1), 50% at $S/N = 7.72^{+0.15}_{-0.14}$. Orange: long-period (HZ-band) regime (periods inside the conservative HZ band of the host cell, 75 planets), 50% at $S/N = 8.61^{+0.33}_{-0.28}$. Bands are bootstrap 1σ intervals of the fitted Γ cumulative distribution function (CDF).

$S/N = 7.72^{+0.15}_{-0.14}$ and exceeds 90% above $S/N \approx 9$ (94%, 165 of 176, at $S/N \geq 9$; 96%, 124 of 129, at ≥ 10). The 50% point falls near the canonical $S/N \approx 7$ detection threshold of the transit literature, here measured end to end on correlated noise.

Restricted to planets whose periods fall inside the conservative HZ band of their host cell (Kopparapu et al. 2013), the curve shifts by $0.9^{+0.8}_{-0.5}$ in S/N (50% at $8.61^{+0.33}_{-0.28}$, $p = 0.012$ from a permutation test reassigning the HZ labels at random and refitting the shift). These injections still carry many transits (7–37, median 18); the shift is a property of long-period signals rather than of the HZ, as non-HZ planets with $P \geq 19$ d show the same displacement. It arises because HZ periods give longer transits, which the Gaussian-process detrending and correlated noise erode more strongly. The forecasts of Sect. 4 use this long-period curve.

The false-alarm rate is measured on the noise-only twins (Fig. 2). It falls from $79 \pm 2\%$ to 0.6% (3 of 465) as the S/N_{pink} floor rises from 6 to 8, while completeness decreases from $61 \pm 2\%$ to $51 \pm 2\%$: noise is confined below $S/N_{\text{pink}} \approx 7.5$ –8 whereas injected planets extend above. A floor at 7.5 yields a $6.0 \pm 1.1\%$ false-alarm rate at $54 \pm 2\%$ completeness. Counting only candidates with a coherent-clock verdict from the event-time test, the rate is 1.9% (9 of 465) for the frozen chain and 1.5% (7 of 465) with the power gate. On correlated noise, part of the apparent completeness at low floors consists of false alarms; the two quantities are therefore reported as a pair at each operating point.

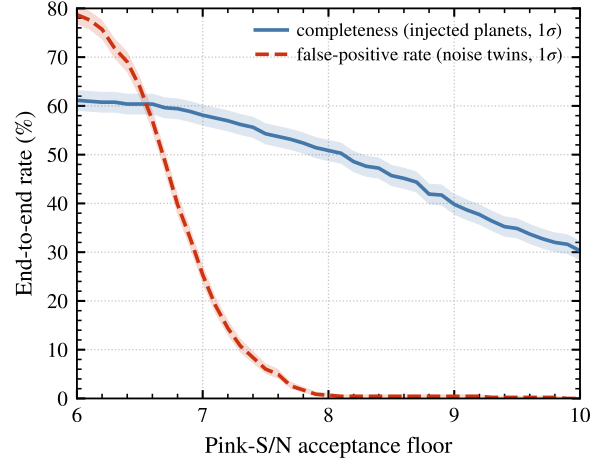


Fig. 2. End-to-end completeness (blue) and noise-twin false-alarm rate (red, dashed) versus the S/N_{pink} acceptance floor, with Wilson 1σ bands (no-systematics arm). The false-alarm rate falls from $79 \pm 2\%$ at a floor of 6 to 0.6% at 8; completeness decreases from $61 \pm 2\%$ to $51 \pm 2\%$ over the same range.

4 Application to the LOPS2 field

The forecast separates measured from modelled quantities. The completeness curves are measured on the campaign cells. Each real star then requires only the placement of its (real or hypothetical) planets on the S/N axis, using the same S/N convention as the campaign. The per-point noise is interpolated between the three measured cells for the faint half of the sample and extrapolated brightward for the rest (interpolation residuals below 3%); stellar parameters derive from Gaia colours calibrated on the 22 field hosts with ExoFOP parameters; and the camera count of each star follows from the four-group field geometry, validated against the published field (union area reproduced to 0.05%; 13/11/33/43% of the field covered by 24/18/12/6 cameras).

Three applications follow. First, the objects already known: the 50 transiting TESS objects of interest around the 36 M-dwarf hosts inside the field footprint, 19 of them confirmed planets and 31 still candidates in the 2026-06-29 ExoFOP snapshot, all have predicted recovery probabilities at unity (median predicted $S/N = 96$); once PLATO data exist, this population provides an on-sky validation sample with known ephemerides. Second, the HZ benchmarks (Fig. 3): a transiting $1.0 R_{\oplus}$ planet receiving conservative-HZ irradiation around the same real stars would be recovered as a candidate with an integrated probability of $41 \pm 1\%$ (statistical only; see below), rising to $65 \pm 1\%$ at $1.5 R_{\oplus}$. The bright cohort of the field (L 98-59, TOI-270, and TOI-700 are members) is recovered at either radius; the faintest hosts are out of reach at any floor. This estimate carries one extrapolation: the completeness curve is measured up to $P = 100$ d, whereas about a third of the real HZ hosts reach 100–160 d, only 4–6 transits over the baseline. That is the genuine few-transit regime the campaign did not sample, where the long-duration argument above no

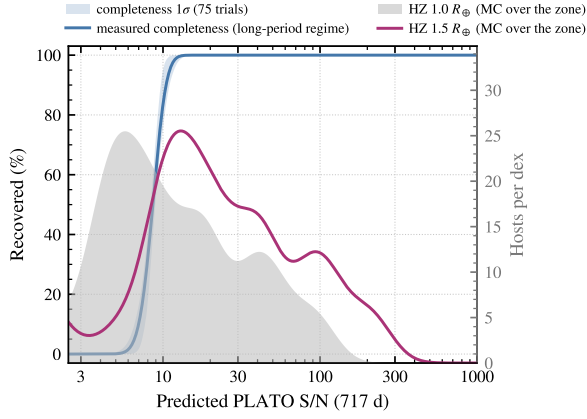


Fig. 3. Measured long-period completeness (blue, with its bootstrap 1σ band; left axis) applied to the real field population (no-systematics arm). Right axis, hosts per dex: predicted PLATO S/N distribution of habitable-zone planets of $1.0 R_{\oplus}$ (grey fill) and $1.5 R_{\oplus}$ (purple), Monte Carlo sampled over the conservative HZ of each of the 35 LOPS2 M-dwarf TOI hosts (of the 36 hosts, the 35 with a catalogued stellar radius), with per-star camera counts. The integrated detection probabilities quoted in the text weight each population by the completeness curve.

longer holds, so the forecast is optimistic for that third. Third, the field-population scale: over the 16 929 M dwarfs of the PLATO input catalogue (Montalto et al. 2021) inside the footprint, with conservative-HZ occurrence rates (Dressing & Charbonneau 2015), about 19 transiting HZ Earths ($1\text{--}1.5 R_{\oplus}$; $10\text{--}38$ over the occurrence-rate interval) are statistically present, of which the pipeline would recover $4.0^{+4.2}_{-1.8}$ as candidates (68% interval propagated from the occurrence rates), plus $6.6^{+5.4}_{-2.8}$ HZ super-Earths ($1.5\text{--}2 R_{\oplus}$). That denominator is a population count inside the footprint, not the list of stars P4 will observe, and the yield scales linearly with the observed fraction. Most of the reservoir lies at $S/N = 2\text{--}6$ around stars too faint for any method. These are candidate counts and carry the reliability of Sect. 3: $6.0 \pm 1.1\%$ false alarms at the adopted floor, $1.5\text{--}1.9\%$ once a coherent-clock verdict is required. For comparison, the consortium performance assessment (Cabrera et al. 2026) quotes ten planets below $2 R_{\oplus}$ in the P4 habitable zone at an intermediate occurrence rate, from sensitivities (Eschen et al. 2024) stated there for the P1, P5 and Prime samples, against the 10.6 recovered here on different stellar and occurrence assumptions.

5 Limitations

The completeness, reliability, and yields of Sects. 3 and 4 are the no-instrument-systematics bound: PSLS stellar and random noise with a perfectly corrected instrument. The systematics arm, matched to it but with the pre-launch model of the post-correction residuals injected, measures the cost of that assumption (Fig. 4): the 50% completeness point shifts by $+0.54^{+0.27}_{-0.22}$ in S/N (7.72 to 8.26), a loss localised on the transition (peaking at about 14 points near $S/N = 10$) and vanishing on the plateau.

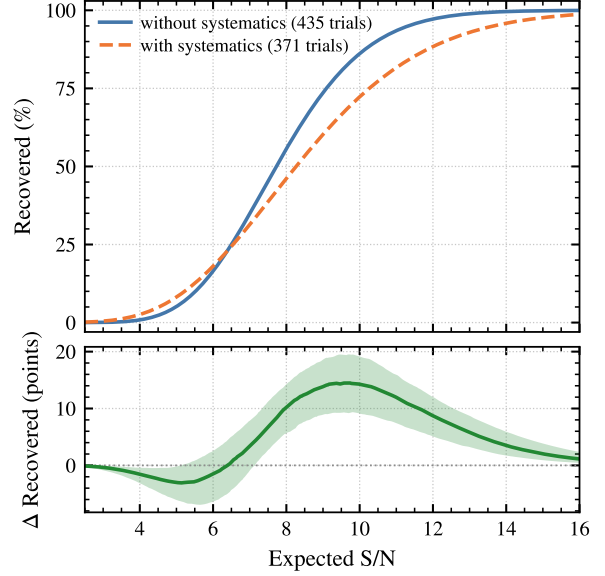


Fig. 4. The systematics penalty. *Top*: end-to-end completeness without (blue) and with (orange, dashed) the modelled instrument systematics, both as Γ -CDF fits. *Bottom*: their difference in recovered percentage points, with its bootstrap 1σ interval.

The penalty is not uniform. The residual template is transplanted at fixed amplitude from the single configuration for which official tables exist: the tables stop at magnitude 13, short of the campaign’s cells at $V = 14.5\text{--}16$, where the random-noise term dominates the error budget. Its effect therefore scales with the amplitude-to-noise ratio of each star: $+1.5$ at 2.9σ on the quietest cell (1308 ppm), $+0.6$ at 1.2σ on the intermediate one, and consistent with zero (-0.5 at 1.1σ) on the noisiest (6200 ppm); the quietest-to-noisiest contrast is itself significant at 3.0σ . This gradient is a property of the injection, not a prediction of the true per-star penalty, which would need an amplitude-versus-magnitude law the tables do not provide; on faint stars the real residuals worsen with charge-transfer inefficiency and would likely reverse the near-zero point.

The acceptance floor is robust. At 7.5 the false-alarm rate is $3.5 \pm 0.9\%$ (below the no-systematics 6%, as the residuals push noise candidates below the floor) for $46 \pm 2\%$ completeness. The HZ-Earth yield is unchanged within the reproducibility of the estimate: the long-period injections that feed it lie in the low-penalty regime, and the arm B completeness curve there is statistically identical to that of arm A. Two omissions push the penalty in opposite directions: the pipeline has no dedicated systematics-correction stage, which overstates the cost, while the residuals are treated as camera-independent, which understates it (common-mode terms correlate within a group). The $+0.54$ is therefore an estimate at the modelled residual level, not a one-sided bound.

Four fidelity limits of PSLS (Samadi et al. 2019) apply. (i) Its stellar models are solar-like; at the campaign’s faint magnitudes the budget is photon- and instrument-dominated, and stellar activity is a fixed spot parameteri-

sation. Flares were not simulated (Sect. 2), which makes completeness and false-alarm rate alike optimistic by an amount no control arm bounds here. Yaptangco et al. (2025), applying the same family of chain to real TESS photometry of M dwarfs, find the 50% threshold moving from $S/N = 7 \pm 1$ on inactive hosts to 14 ± 1.5 on active ones: the present completeness describes the quiet end of the population. (ii) Its activity is one frozen spot configuration, the same three spots of 75, 90, and 60 d lifetime for every star, so the activity-driven false-alarm channel is sampled at one point of the distribution, not bounded. (iii) Its systematics are the residuals left after the planned mission corrections (drift correction and microscan assumed applied; the end-of-life charge-transfer-inefficiency correction not yet quantified), treated here as independent between cameras, whereas the dominant common-mode terms such as pointing jitter are correlated within a camera group (Jannsen et al. 2024), so the real multicamera averaging is less favourable. (iv) Pixel-level effects and blends are out of scope, so blended astrophysical false positives are not probed.

The forecast’s model limits are: (1) the occurrence rates dominate the field-population interval (a factor of two); (2) the noise interpolation, anchored at P -band magnitudes 13.5–15.0, is extrapolated brightward for 15 of the 35 HZ hosts, which carry about 80% of the integrated probability, most of them on the completeness plateau; (3) activity is fixed at the cells’ median level; (4) the 18- and 24-camera zones (24% of the field) use a $1/\sqrt{N_{\text{cam}}}$ scaling valid only when photon-limited; (5) the HZ forecast extends the completeness curve beyond its measured $P \leq 100$ d domain for about a third of the hosts (Sect. 4); (6) the limb-darkening window factor is held constant over impact parameter (-1.7 points on the $1.0 R_{\oplus}$ forecast); (7) binaries are unfiltered and orbits circular; (8) the coherence test’s per-event timing degrades for transits longer than about 4 h, a PLATO regime inherited from TESS and deferred to a future version; and (9) the injections sample impact parameters uniformly over 0–0.6 whereas the forecast draws them over 0–0.9, so about a third of the forecast population lies outside the range on which the completeness was measured, in the unfavourable direction. The quoted $41 \pm 1\%$ reflects the completeness-curve statistical width alone; the model limits above point in both directions, the camera scaling, the period extrapolation and the brightward noise extrapolation making it an over-estimate, without their relative weights being established here. The completeness and yield are scored at the frozen pink- S/N floor of 7; at the adopted discovery floor of 7.5 they are a few points lower.

6 Conclusions

An end-to-end injection-recovery campaign on simulated PLATO photometry of M dwarfs yields three measured results. First, a pipeline validated on TESS recovers 50% of injected planets at an expected $S/N = 7.7 \pm 0.15$, and the long-period regime that contains the habitable zones is measurably harder (8.6 ± 0.3), so an HZ forecast uses the completeness curve of the corresponding period regime.

Second, the false-alarm rate on planet-free light curves is a strong function of the red-noise-aware acceptance floor, falling from $79 \pm 2\%$ to 0.6% between floors of 6 and 8; completeness and reliability are therefore only meaningful as a pair, and the measured pair motivates an operating floor of $S/N_{\text{pink}} \approx 7.5$ for discovery ($6 \pm 1\%$ false alarms at $54 \pm 2\%$ completeness) or 8 for statistical purity (0.6%), floors that the systematics arm leaves unchanged. Third, applied to the real LOPS2 M dwarfs, the known transiting TESS objects of interest of the field would all be recovered, while a transiting habitable-zone Earth would be recovered as a candidate about two in five times ($\sim 40\%$), placing these planets at the detection frontier, where gains in pipeline completeness map directly onto the expected number of candidates; both deciding quantities can be measured, as done here, before launch.

Data availability

The scored per-planet and per-twin tables of both arms, the generator scripts with the PSLS configurations and random seeds that regenerate the 1720 light curves and the residual templates of the systematics arm, and the scripts that produce the figures are available at <https://gitlab.com/yohannschudi/plato-p4-completeness>, together with the detection chain at the frozen version used for the campaign. PSLS is distributed by its authors (Samadi et al. 2019). The LOPS2 field population was drawn from the all-sky PLATO input catalogue (Montalto et al. 2021) through Vizier.

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